NEWSPAPER

ITEM WISE EDITORIALS

2nd May, 2019

18th WEEK

ERADICATION OF POLIO & PAKISTAN

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... DISCUSSION

2nd MAY, 2019

EDITORIAL DISCUSSION

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Fighting polio in Pakistan

Instead of insisting on the oral polio vaccine, using the inactivated polio vaccine along with other vaccines will help

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THE HINDU

2nd May, 2019

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Science and Technology - developments and their applications and effects in everyday life
Achievements of Indians in science & technology; indigenization of technology and developing new technology.

3 Fighting polio in Pakistan

Instead of insisting on the oral polio vaccine, using the inactivated polio vaccine along with other vaccines would help

Last month, the polio eradication programme in Pakistan was in the news for all the wrong reasons. On April 22, a government hospital in Mashokhel in Peshawar district was set on fire after many children allegedly fell sick after being given the anti-polio vaccine. On April 23 and 24, in two separate incidents, two police officers guarding vaccinators were shot dead. On April 25, in Chaman, which borders Afghanistan, a polio worker was shot dead and her helper injured. Since December 2012, nearly 90 people have been killed in the country for working to eradicate polio. Due to recurrent threats to workers, the Pakistan government has now suspended the anti-polio drive.

Cases of wild poliovirus type 1

This is the worst time to take this decision. This year alone, eight paralysed children with wild poliovirus type 1 (WPV1) have been found in Pakistan. Environmental surveillance by testing sewage samples has shown 91 WPV1-positive samples, in the provinces of Khyber Pakhtunkhwa, Balochistan, Punjab and Sindh. In the past week alone, 13 sewage samples were found to be positive for WPV1.

This is a worrying sign. With suspended immunisation activities, WPV1 will spread fast and the number of polio cases could increase and cause an outbreak. If Pakistan cannot eliminate polio, the global eradication programme is sure to stall. When India eliminated WPV1 in January 2011, the Global Polio Eradication Initiative (GPEI), a public-private partnership led by national governments with five partners, did not ask if Pakistan would be able to follow suit; it simply assumed it would. This was unrealistic. In Uttar Pradesh and Bihar, oral polio vaccine (OPV) coverage of 98-99% was sustained with an average of 15 doses per child from 2003. There was full cooperation from the health workers and the public. The war on polio requires such intensity and coverage and it is unrealistic to expect this in Pakistan, where polio eradication is falsely depicted as a Western agenda with the sinister motive of reducing fertility.

...Contd.

3 Fighting polio in Pakistan

The GPEI has pinned all its hopes on the OPV and has excluded the alternate inactivated polio vaccine (IPV) to eradicate WPVs. The OPV is cheap and easy to give to children, but it has to be given to them again and again in pulse campaigns since its efficacy is poor. On the other hand, the IPV is highly efficacious and needs to be given just two-three times as part of routine immunisation.

Risk of polio outbreaks

The OPV has another problem. If coverage declines (as is bound to happen in Pakistan), vaccine viruses will spread to children who are not vaccinated, back-mutate, de-attenuate and become virulent. Such viruses are called circulating vaccine-derived polioviruses (cVDPV). They can cause polio outbreaks. Thus Pakistan will soon be at risk of polio outbreaks by both WPV1 and cVDPV.

It is to avoid the emergence of cVDPV that India strives to maintain high OPV coverage through routine immunisation, Mission Indradhanush and annual national pulse campaigns. In 2018, Papua New Guinea developed a cVDPV polio outbreak as OPV coverage fell to 60%. In 2017, as OPV coverage fell to 53%, Syria had an outbreak of cVDPV polio.

There is yet another problem in Pakistan. With the OPV being identified as the weapon in the war on polio and with some in Pakistan believing that the aim of eradication is to reduce fertility, a vaccine is given only three or four times, not 15-20 times.

Hope is not lost for polio eradication provided that the GPEI relents on its insistence on the OPV and uses the IPV along with other common vaccines. IPV-containing vaccines could be included in the routine immunisation programme and given without attracting the attention of militants. The false propaganda about polio vaccination in Pakistan will then lose its sting. While near-100% coverage with the OPV is necessary, 85-90% coverage with the IPV given in a routine schedule would be sufficient.

...Contd.

3 Fighting polio in Pakistan

If the GPEI insists on the OPV as the only weapon against polio, we have hit the end of the road in Pakistan. But the world cannot afford to lose this war on polio. India could show the way forward by giving the IPV in its universal immunisation programme (at least two doses and preferably three) and then discontinuing the infectious OPV altogether.

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EDITORIAL DISCUSSION

Protecting against polio

Why the inactivated polio vaccine is essential for India

R. PRASAD

The VAPP burden

Switching to IPV

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INDIA'S NATIONAL NEWSPAPER SINCE 1878

8th November, 2018

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IMPORTANT QUESTION

ESSAY/ MAINS / GD / INTERVIEWS

Examine the need to switch-over to Inactivated Polio Vaccine, with specific reference to India.

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- Pakistan Government suspended the anti-polio drive following increasing number of attacks on polio workers.
- Countrywide campaign to administer anti-polio drops to 39 million children under five years of age was launched.
- However, the campaign, ran into trouble after reports that several children were taken to hospital as they fell sick
- Pakistan is one of the three countries, where polio is still endemic.
- Since December, 2012, nearly 90 people have been killed in the country for working to eradicate polio.

Pakistan and Afghanistan, along with Nigeria, are the only three countries in the World, where polio remains endemic.

EDITORIAL DISCUSSION

ORAL POLIO VACCINE

1. It contains live weakened virus.
2. OPV was used for eradicating purposes, but, without fully protecting the children.
3. It is easy to administer OPV than IPV.
4. Cost per dose of OPV is also lower than that of IPV.
5. India and low-income countries continued to rely on the OPV.

INACTIVATED POLIO VACCINE

1. It uses dead virus to immunize the children.
2. IPV produces humoral immunity involving antibodies in body fluids, so the immunised child does not get paralysis, but it can't stop the circulation of wild polio viruses.
3. The viruses will continue to circulate in the community.
4. High-income countries preferred IPV.
5. India licensed the IPV only in 2006 but, did not introduce it in routine immunisation.

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EDITORIAL DISCUSSION

PROBLEMS WITH LACK OF ORAL POLIO VACCINE IMMUNISATION AND PROBLEMS

1. It is cheap and easy to administer to children, it has to be given again and again in pulse campaigns since its efficacy is poor.
2. If coverage declines, vaccine viruses will spread to children, who are not vaccinated. It will back-mutate and becomes virulent.
3. Such viruses are called circulating Vaccine-Derived Polioviruses (cVDPV). This can cause polio outbreaks.
4. Pakistan will soon be at the risk of polio outbreaks by both WPV1 and cVDPV.

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MORE ON CIRCULATING VACCINE DERIVED POLIOVIRUSES

1. In 2018, Papua New Guinea developed a cVDPV polio outbreak as OPV coverage fell to 60%.
2. In 2017, as OPV coverage fell to 53%, Syria had an outbreak of cVDPV polio.
3. That is the reason India strives to maintain high OPV coverage through routine immunisation i.e., Mission Indradhanush and Annual National Pulse Campaigns.

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THE NEED TO INTRODUCE INACTIVATED POLIO VACCINE

1 Inactivated Polio Vaccines are highly efficacious, they need to be given just two-three times as part of routine immunisation.

2 IPV can be used along with other common vaccines.

3 IPV containing vaccines could be included in the routine immunisation programme, they can be given without attracting the attention of militants.

4 The false propaganda about polio vaccination will then vanish.

5 100% coverage with OPV is necessary, but, 85-90% coverage with IPV would be sufficient to prevent outbreaks.

India should also consider giving IPV in its Universal Immunisation Programme and then discontinuing the infectious OPV altogether.

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